

Assignment 4

Debate on ethical issues in Biotechnology

Biotechnology is moving faster than our legal and ethical frameworks can keep up with. In this assignment, you will step out of the laboratory and into the boardroom, the courtroom, or the community.

In this assignment, you will read about the assigned topics, gather information and debate about the ethical, legal and social issues arising from the biotechnological advances in given topics.

The class will be divided into 6 groups of 6 members each. Each group will tackle one of the following dilemmas:

Group	Debating theme
1	Human Germline Editing (CRISPR) Should we permit the permanent alteration of the human genome to eliminate disease and enhance traits?
2	De-Extinction and "Resurrection Biology" Should we use cloning and gene-splicing to bring back species like the Woolly Mammoth or Passenger Pigeon?
3	Genetic Privacy and Law Enforcement Should law enforcement be allowed to use consumer DNA databases (Ancestry/23andMe) to track down suspects via their relatives?
4	Genetically modified organisms and gene-edited crops GM technology remains socially controversial despite extensive scientific evidence supporting the safety of many approved GM products. Should the world embrace GMOs in food?
5	Genetic engineering of animals and their welfare Is genetically engineering animals to increase productivity, resist disease, or fit intensive farming systems for human benefit ethically justifiable, or does it reduce sentient beings to mere industrial commodities?

6	Synthetic Biology Should humans be allowed to design and synthesize entirely new organisms from scratch?
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Debate

- Each group should be divided into 2 teams of 3 members to debate the topic in following views:
 - ❖ Affirmative
 - ❖ Negative/opposing
- **Group members can discuss and divide themselves into affirmative or opposing roles.**
- Team members may assume pretend roles relevant to the case discussed (eg: Scientists, Professors, Detectives, Activists, Ethicists...)
- A moderator must be appointed from another group. They should introduce the team and initiate the debate
- A timekeeper too must be appointed from another group to keep the time and ensure speakers do not exceed their allocated time.

Time allocation:

1st round

1. Affirmative group Leader brings out important positive points – 2 minutes
2. Opposing team Leader brings out important negative points – 2 minutes
3. 2 members speak 2 minutes first from Affirmative and then from opposing team alternatively.

2nd round

1. Affirmative team can present their counter arguments – 2 minutes per person
2. Opposing team presents their counter arguments – 2 minutes per person

Final statement

Each Leader presents their summarizing statements at the end - 2 minutes each

Add the photos and the summary of your group debate to your portfolio.

Date of the debate- 2nd September 2026

Evaluation

Knowledge ----- 40

Presentation ----- 50

Time spent ----- 10

Total 100